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## Full Notes: Conditions + SELECT + CASE in MySQL

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### Step 1: Table Create karo — employees

```
CREATE TABLE employees (  
  id INT PRIMARY KEY,  
  name VARCHAR(50),  
  department VARCHAR(50),  
  salary INT,  
  city VARCHAR(50)  
);
```

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### Step 2: Data Insert karo

```
INSERT INTO employees (id, name, department, salary, city) VALUES  
(1, 'Rahul', 'IT', 60000, 'Delhi'),  
(2, 'Priya', 'HR', 45000, 'Mumbai'),  
(3, 'Aman', 'Finance', 72000, 'Delhi'),  
(4, 'Neha', 'IT', 50000, 'Pune'),  
(5, 'Ravi', 'HR', 30000, 'Chennai');
```

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### Step 3: Basic SELECT + WHERE + AND + OR Conditions

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#### Example 1: WHERE with AND

```
SELECT name, department, salary, city  
FROM employees  
WHERE department = 'IT' AND city = 'Delhi';
```

#### Output:

| name | department | salary | city |
|------|------------|--------|------|
|------|------------|--------|------|

|       |    |       |       |
|-------|----|-------|-------|
| Rahul | IT | 60000 | Delhi |
|-------|----|-------|-------|

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#### Example 2: WHERE with OR

```
SELECT name, department, salary, city  
FROM employees  
WHERE city = 'Mumbai' OR city = 'Pune';
```

#### Output:

| name | department | salary | city |
|------|------------|--------|------|
|------|------------|--------|------|

|       |    |       |        |
|-------|----|-------|--------|
| Priya | HR | 45000 | Mumbai |
|-------|----|-------|--------|

| name | department | salary | city |
|------|------------|--------|------|
|------|------------|--------|------|

|      |    |       |      |
|------|----|-------|------|
| Neha | IT | 50000 | Pune |
|------|----|-------|------|

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### Example 3: Complex WHERE with AND + OR

```
SELECT name, department, salary, city
FROM employees
WHERE (department = 'HR' AND salary > 40000) OR city = 'Pune';
```

#### Output:

| name | department | salary | city |
|------|------------|--------|------|
|------|------------|--------|------|

|       |    |       |        |
|-------|----|-------|--------|
| Priya | HR | 45000 | Mumbai |
|-------|----|-------|--------|

|      |    |       |      |
|------|----|-------|------|
| Neha | IT | 50000 | Pune |
|------|----|-------|------|

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### Step 4: SELECT + CASE — Salary Status

```
SELECT name, salary,
CASE
    WHEN salary >= 70000 THEN 'High Salary'
    WHEN salary >= 50000 THEN 'Medium Salary'
    ELSE 'Low Salary'
END AS salary_status
FROM employees;
```

#### Output:

| name | salary | salary_status |
|------|--------|---------------|
|------|--------|---------------|

|       |       |               |
|-------|-------|---------------|
| Rahul | 60000 | Medium Salary |
|-------|-------|---------------|

|       |       |            |
|-------|-------|------------|
| Priya | 45000 | Low Salary |
|-------|-------|------------|

|      |       |             |
|------|-------|-------------|
| Aman | 72000 | High Salary |
|------|-------|-------------|

|      |       |               |
|------|-------|---------------|
| Neha | 50000 | Medium Salary |
|------|-------|---------------|

|      |       |            |
|------|-------|------------|
| Ravi | 30000 | Low Salary |
|------|-------|------------|

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### Step 5: SELECT + CASE + WHERE + AND + OR Together

### Example: Salary status for only IT or HR employees living in Delhi or Mumbai

```

SELECT name, department, city, salary,
       CASE
         WHEN salary >= 70000 THEN 'High Salary'
         WHEN salary >= 50000 THEN 'Medium Salary'
         ELSE 'Low Salary'
       END AS salary_status
FROM employees
WHERE (department = 'IT' OR department = 'HR')
      AND (city = 'Delhi' OR city = 'Mumbai');

```

### Output:

| name  | department | city   | salary | salary_status |
|-------|------------|--------|--------|---------------|
| Rahul | IT         | Delhi  | 60000  | Medium Salary |
| Priya | HR         | Mumbai | 45000  | Low Salary    |

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### Summary Table: SQL Conditions + CASE

| Clause | Use Case                                | Example                                              |
|--------|-----------------------------------------|------------------------------------------------------|
| WHERE  | Filter rows based on condition          | WHERE salary > 50000                                 |
| AND    | Multiple conditions true simultaneously | WHERE city='Delhi' AND department='IT'               |
| OR     | Either condition true                   | WHERE city='Mumbai' OR city='Pune'                   |
| CASE   | Create conditional columns              | CASE WHEN salary>70000 THEN 'High' ELSE 'Low'<br>END |

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